Level 2

Control Flow & Loops

If-else, For loops, While loops, Logical Operators

Question1: Write a script that asks for a password and only allows access if it matches "s3cr3t".

```
password = input("Enter the password: ")
if password == "s3cr3t":
    print("Access granted.")
else:
    print("Access denied.")
```

```
(base) nadaabuzaid@Nadas-MacBook Level2 % python3 1.py
Enter the password: s3cr3ttt
Access denied.
(base) nadaabuzaid@Nadas-MacBook Level2 % python3 1.py
Enter the password: s3cr3t
Access granted.
```

Question2: Print all numbers from 1 to 100 except numbers divisible by 4.

```
for i in range(1, 101):
    if i % 4 != 0:
        print(i)
```

```
(base) nadaabuzaid@Nadas-MacBook Level2 % python3 2.py
1
2
3
5
6
7
9
10
11
13
14
15
17
18
```

Question3: Print every 4-digit PIN code (0000–9999).

```
for pin in range(10000):
    print(f"{pin:04d}")
```

```
(base) nadaabuzaid@Nadas-MacBook Level2 % python3 3.py
0146
0147
0148
0149
0150
0151
0152
0153
0154
0155
0156
```

Question4: Check if a given year is a leap year.

```
year = int(input("Enter a year: "))
if(year % 400 == 0 or (year % 100 != 0 and year % 4 == 0)):
    print(f"{year} is a leap year.")
else:
    print(f"{year} is not a leap year.")
```

```
(base) nadaabuzaid@Nadas-MacBook Level2 % python3 4.py
Enter a year: 2001
2001 is not a leap year.
(base) nadaabuzaid@Nadas-MacBook Level2 % python3 4.py
Enter a year: 2004
2004 is a leap year.
```

Question5: Find prime numbers between 1 and 100.

```
for i in range(2, 101):
    is_prime = True
    for j in range(2, i):
        if i % j == 0:
            is_prime = False
            break
    if is_prime:
        print(i)
```

```
(base) nadaabuzaid@Nadas-MacBook Level2 % python3 5.py
2
3
5
7
11
13
17
19
23
29
31
37
41
43
47
53
59
61
67
71
73
79
83
89
97
```

Question6: Simulate a login system that locks after 3 failed attempts.

```
failed_attempts = 0
password = "secret"

while failed_attempts < 3:
    entered_password = input("Enter the password: ")
    if entered_password == password:
        print("Access granted.")
        break
    else:
        failed_attempts += 1
        print(f"Access denied. Attempt {failed_attempts}/3.")
```

```
(base) nadaabuzaid@Nadas-MacBook Level2 % python3 6.py
Enter the password: nada
Access denied. Attempt 1/3.
Enter the password: nada
Access denied. Attempt 2/3.
Enter the password: n333
Access denied. Attempt 3/3.
(base) nadaabuzaid@Nadas-MacBook Level2 % python3 6.py
Enter the password: nada
Access denied. Attempt 1/3.
Enter the password: secet
Access denied. Attempt 2/3.
Enter the password: secret
Access granted.
```

Question7: Create a number guessing game.

```
secret_num = 42
while True:
    guess = int(input("Guess the number (between 1 and 100):
"))
    if guess == secret_num:
        print("Congratulations! You guessed the number.")
        break
    elif guess < secret_num:
        print("Too low. Try again.")
    else:
        print("Too high. Try again.")
```

```
(base) nadaabuzaid@Nadas-MacBook Level2 % python3 7.py
Guess the number (between 1 and 100): 20
Too low. Try again.
Guess the number (between 1 and 100): 53
Too high. Try again.
Guess the number (between 1 and 100): 42
Congratulations! You guessed the number.
```

Question8: Print numbers 1-100, replacing multiples of 3 with "Fizz", 5 with "Buzz", both with "FizzBuzz".

```
for i in range(1, 101):
    if i % 3 == 0 and i % 5 == 0:
        print("FizzBuzz")
    elif i % 3 == 0:
        print("Fizz")
    elif i % 5 == 0:
        print("Buzz")
    else:
        print(i)
```

```
(base) nadaabuzaid@Nadas-MacBook Level2 % python3 8.py
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
```

Question9: Continuously ask for a password until the correct one is entered.

```
password = ""

while password != "secret":
    password = input("Enter the password: ")

print("Access granted.")
```

```
(base) nadaabuzaid@Nadas-MacBook Level2 % python3 9.py
Enter the password: test
Enter the password: pass
Enter the password: 1234
Enter the password: se
Enter the password: secet
Enter the password: secret
Access granted.
```

Question10: Check if a string is a palindrome.

```
text = input("Enter a string: ")
if text == text[::-1]:
    print(f"{text} is a palindrome.")
else:
    print(f"{text} is not a palindrome.")
```

```
(base) nadaabuzaid@Nadas-MacBook Level2 % python3 10.py
Enter a string: banasn
banasn is not a palindrome.
(base) nadaabuzaid@Nadas-MacBook Level2 % python3 10.py
Enter a string: madam
madam is a palindrome.
```